Round 7 (Astra): The Whole Error Is Three Numbers per Neuron

How to compute variance and kappa_4 diagonal corrections more accurately, at low FLOP cost

Width 1024, depth 16, bias-free ReLU MLP | Baseline: public V29 factorized K=3 chain

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Executive conclusion

The new measurements change the optimization target. The dominant problem is no longer "represent the whole distribution better." The mean closure is already essentially exact when fed the correct marginal variance, kappa_3 diagonal, and kappa_4 diagonal. Therefore the best next estimator should spend FLOPs only on producing those scalar channels more accurately.

My ranking is: (1) fix the variance channel with a deeper bivariate-only Edgeworth/Wick closure using the pairwise slices V29 already has; (2) compute the missing kappa_4 diagonal through one-step directional ReLU-birth contractions, not through a persistent fourth-order tensor; (3) use sampling only as a falsification tool unless a sample-level control variate achieves R^2 above about 0.99 for squared observables. I do not see a credible 100x sampling variance reduction from the current chain state.

The highest-priority experiment is the variance "pair super-closure." It is almost free in n^3 units and directly tests whether the fresh ~4.2e-5 covariance error is merely the next omitted connected bivariate diagrams. If that works, it can recover a large fraction of the 47% variance-oracle gain without touching the expensive K3 source machinery.

Channel / proposal	Persistent state	Incremental cost	Oracle value	Priority
Variance: bivariate super-closure	No new tensor; existing C, D21, K31, K22, marginals	~0.05-0.20 u/layer; likely 1-3 u/network	Raw MSE 2.290e-8 -> 1.220e-8	1
kappa_4: directional fresh-birth hypertrees	Temporary n x m message matrices only	~1.2-1.6 u/layer for first 1111 test at m=300-400; late-layer version ~7-10 u/network	Raw MSE -> 1.600e-8 oracle	2
kappa_4: add 211 hypertree messages	Same temporary messages; uses C and D21	~additional 1-3 u/layer depending message reuse	Potentially captures missing pattern	3
Correction-only sampling	Sample batch + optional affine Gaussian surrogate	0.016 u/sample before CV overhead	All channels in principle	4 / diagnostic only

Cost convention

One unit u is one dense n by n matrix product at n=1024, approximately 2 n^3 = 2^31 FLOPs. Therefore an n by n times n by m product costs approximately (m/n) u. For m=300, 350, 400 this is 0.293, 0.342, 0.391 u. A single elementwise n^2 pass costs about 1/(2n) = 0.000488 u.

Q1 - Variance channel

Key point: do not remove the full covariance state first. In V29 the covariance machinery is a small fraction of the bill, and C_off is also consumed by K3 and kappa_4 births. The opportunity is to make the nonlinear C_post closure more accurate using only O(n^2) work, then let the existing covariance sandwich propagate that correction automatically.

1. Proposed method: a bivariate-only "super-closure"

For each pair (i,j), Cov(ReLU(z_i), ReLU(z_j)) depends only on the bivariate law of (z_i,z_j). This is crucial: once the state already contains the pairwise cumulant slices C_ij, K3_iij / K3_ijj, K4_iiij / K4_ijjj, K4_iijj, and the marginals, every connected diagram built only from those blocks is an elementwise pair calculation. It does not require a tensor contraction or any n^3 operation.

V29 evaluates a pruned finite diagram table. The next experiment should regenerate the bivariate table with a larger diagram-cost budget, while forbidding any block of degree > 4. This adds products of cumulants V29 already knows; it does not add a K5 or K6 state.

> C_post(i,j) = sum_{connected bivariate diagrams Gamma} c_Gamma * Product[known pair slices] * Product[ReLU Wick coefficients].
>
> Keep only diagrams whose individual cumulant blocks have degree <= 4; increase the total diagram budget from the current truncation to cost 4 and then cost 5.

A direct enumeration of connected two-group vector partitions using only cumulant blocks of degree <=4 gives 26 symmetry classes through cost 4 and 55 through cost 5 (10 through cost 3). Even a few hundred n^2 elementwise operations are cheap: 300 n^2 operations are only about 0.146 u per layer before fusion.

Why this is the best explanation to test first

- The fresh covariance error is approximately stationary across depth, which is the signature of a local nonlinearity truncation rather than transport drift alone.
- The output covariance is bivariate, so higher products of already-available pair cumulants can be evaluated without the Khatri-Rao wall that hurts K3 transport.
- A 2% per-entry residual in C_off is entirely compatible with one additional 1/sqrt(n)-suppressed diagram rung at n=1024.
- If the experiment fails with teacher-forced K<=4 pair slices, then the result is decisive: the missing variance information lives in genuine K5/K6 bivariate cumulants or in errors of the supplied slices, not in a missing cheap product diagram.

2. Falsifier on the stored arrays

1. For every layer, reconstruct the V29 post-ReLU covariance using the true pre-activation marginals and the true stored D21, K31, K22, and K4 diagonal. Keep V29 coefficients unchanged. This isolates input-slice error from closure-truncation error.
2. Regenerate the connected bivariate diagram table at cost 4, but permit only degree-2/3/4 cumulant blocks that exist in the stored arrays. Evaluate all pairs.
3. Repeat at cost 5.
4. For each version compute: fresh C_post residual RMS; correlation with the true fresh residual; and, more importantly, the projected next-layer variance residual diag(W dC W^T) on neurons with |mu/sigma| < 0.75.

Kill rule: if teacher-forced cost-5 reduces the projected variance error by less than 25%, stop. If cost-4 or cost-5 removes at least half of the projected error, this becomes the first production port.

3. FLOPs and state

Variant	Extra state	Extra per-layer work	Network-scale estimate
Cost-4 pair super-closure	None beyond existing pair slices	roughly 50-150 n^2 elementwise ops	~0.4-1.1 u over 15 nonlinear layers
Cost-5 pair super-closure	None	roughly 150-400 n^2 ops depending fusion	~1-3 u over 15 layers
Target-row quadratic form only	m outputs	one n x n by n x m multiply	m/n u per corrected layer = 0.29-0.39 u for m=300-400

Because V29 already forms and propagates C, I would replace the existing C_post closure with the improved one rather than maintain a separate shadow covariance. That automatically carries the inherited correction forward at no additional n^3 cost.

4. Why pair screening is unlikely to work

For a next-layer random row q, the variance correction is q^T dC q. With q_j approximately N(0,2/n), the weights q_j q_k have no stable heavy tail. Unless the teacher residual dC itself has sparse pair support, selecting a small fraction of pairs loses cancellation-sensitive content. The under-one-hour falsifier is simple: on the stored true dC, sort |q_j q_k dC_jk| and measure how many pairs are required for 99% of the quadratic-form energy. If more than about 20% of pairs are needed, pair screening cannot beat an n^2 pair closure.

Q2 - kappa_4 diagonal

The right representation is directional, one layer at a time. The measurements say the deep fourth cumulant is regenerated at every ReLU and that a transported old "star" explains only about 10% of the deep diagonal. Therefore do not carry a full K4 tensor. Compute the missing fresh ReLU birth, contract it immediately with the next weight row, and add the resulting scalar directly to the next pre-activation kappa_4 diagonal.

1. All-distinct 1111 birth: exact leading hypertree algebra

ARC's public factor_k4 implementation explicitly groups the leading connected all-distinct diagrams into three topologies: path, star, and a K3+C "3+2" hypertree. Its combinatorial coefficients are 12, 4, and 12. Contracting all four output legs with the same next-layer row q collapses those tensors to messages.

Let C be the pre-activation off-diagonal covariance, K3 the pre-activation third cumulant, and a_r = w(r,1) the ReLU Wick coefficients. Define:

x = q .* a1

r = C x

s = q .* a2 .* r

t_j = sum_{k,l} K3[j,k,l] x_k x_l.

Then the leading fresh all-distinct fourth-cumulant contraction is:

Delta_k4_1111(q) ~= 4 * sum_j (q_j a3_j) r_j^3
+ 12 * s^T C s
+ 12 * sum_j s_j t_j.

Terms: star path K3 + C (3+2)

This is the same path/star/3+2 decomposition used in the reference factor-K4 code, but evaluated only in the one direction q that the next layer actually needs. No n^4 object is created.

2. Cheap approximation to the K3 directional contraction

The expensive piece above is t = K3[:,x,x]. Before paying for all source factors, test whether its repeated-index part is sufficient after a random dense directional contraction. Using D21[a,c] = K3[a,a,c] and D3[a] = K3[a,a,a], the exact contribution from all terms with at least two equal K3 indices is:

t_rep = D21^T (x .* x) + 2 x .* (D21 x) - 2 D3 .* (x .* x).

The subtraction removes the triple-counting of K3[j,j,j]. For a batch X of m directions, t_rep needs only two n by n times n by m products. Therefore the full star + path + slice-K3 1111 correction costs about 4(m/n) u per layer.

m active next neurons	4(m/n) u / layer	15-layer worst case	6-layer late-only test
300	1.17 u	17.6 u	7.0 u
350	1.37 u	20.5 u	8.2 u
400	1.56 u	23.4 u	9.4 u

Falsifier: on 64-128 next-layer rows, compute t_exact from the chain's stored factorized K3 sources and compare t_rep. The relevant metric is not ||t_rep-t|| alone; compare the final scalar 12 sum(s .* t). If the weighted scalar has <20% relative RMS error, use the slice form. If it fails, decompose t_exact by K3 source age and test whether the newest one or two sources dominate. A young-only exact t is still much cheaper than a full K4 state.

3. The 211 birth: weighted capped hypertree contraction

For K4[a,a,b,c] with a,b,c distinct, write u_a=q_a^2 and v_b=q_b. The required next-layer contribution is 6 sum_{a,b,c distinct} u_a v_b v_c K211[a,b,c]. The public augmented code already evaluates a related "capped" 211 object by summing over a without materializing n^3. For our purpose, replace the unweighted cap by u and contract b,c with v immediately.

Delta_k4_211(q) = 6 * sum_{a,b,c distinct} q_a^2 q_b q_c * K211_birth[a,b,c].

At the leading lower-order birth rung (C^3 and K3*C), there are 12 symmetry classes for the (2,1,1) output grouping. Nine are hypertrees and can be evaluated by leaf-message elimination in O(n^2) per direction. The three hard classes are: one covariance triangle and two classes in which an all-distinct K3 hyperedge overlaps a covariance edge on two vertices. Those are the non-hypertree remainder.

The cheap nine classes use only matrices that are already present: C, C Hadamard-squared, D21, D21^T, and node-wise Wick vectors w(k,2) on the repeated group and w(k,1) on the single groups. A representative tree contraction is:

sum_{a,b,c} U_a V_b V_c E_ab F_ac = sum_a U_a * (E V)_a * (F V)_a,

where E and F may be C, C.*C, D21, or D21^T after the appropriate Wick row/column scalings.

This is the practical reason to split hypertree from non-hypertree content: the former reduces to reusable n by n times n by m messages; the covariance triangle does not.

4. Q2 falsifier ladder

5. Construct the true missing next-layer K4-diagonal residual using the stored post-ReLU K4 slices: subtract V29's current diagonal + 22 + 31 contribution from the exact row contraction. Restrict scoring to neurons with |mu/sigma| < 1.5.
6. Predict that residual using only the 1111 star term.
7. Add the 1111 path term.
8. Add 3+2 with t_rep. On a subset, compare against 3+2 with exact factorized-K3 t.
9. Add the nine cheap 211 hypertree classes via weighted capped messages.
10. Only if the previous step is strong, measure the three hard non-hypertree classes on a small neuron subset to determine their residual share. Do not implement them end-to-end unless they explain a large fraction of what remains.

Kill rule: if star+path+3+2 does not reduce the relevant K4-diagonal residual RMS by at least 25% or achieve residual correlation above about 0.6, stop before 211. If adding the nine 211 hypertrees does not take total RMS reduction above 40%, this route is unlikely to beat its score cost.

5. End-to-end use

At ReLU layer l, compute the fresh directional correction using rows of W_{l+1}. Add it directly to V29's pre-activation kappa_4 diagonal at layer l+1. Do not create a persistent K4 source. This is a one-step birth correction. The hypothesis is justified by the measured fact that deep K4 is regenerated at every ReLU and old transported K4 contributes only a small fraction of the deep diagonal pattern.

Start with the last 6 layers. With a typical mean-error gain below one per layer, late corrections have the largest final influence. A 6-layer trial also keeps the first directional 1111 patch near 7-10 u for m=300-400, far below the score break-even cost for a substantial fraction of the 30% K4 oracle gain.

Q3 - Sampling the corrections

My conclusion is negative unless a new sample-level control variate is extraordinarily strong. The relevant arithmetic is unforgiving.

Samples	Forward cost at 0.016 u/sample	Chain + samples (starting from 260 u)
10,000	160 u	420 u
20,000	320 u	580 u
30,000	480 u	740 u
40,000	640 u	900 u
~47,000	~752 u	~1012 u

For a roughly Gaussian scalar, the relative standard error of a sample variance is about sqrt(2/N). At N=10,000 this is about 1.4%, while the target variance accuracy is around 0.1%. A correction-only control variate therefore needs roughly 100-200x variance reduction to make 10k samples competitive.

> For an optimal linear control variate: VRF = 1 / (1 - R^2).
>
> 100x VRF requires R^2 = 0.990; 200x requires R^2 = 0.995; 30x still requires R^2 = 0.967.

1. Rao-Blackwellisation over input radius

Because the network is 1-homogeneous, h(Ru)=R h(u). With x~N(0,I), R and u are independent, so E[h^p]=E[R^p] E[h(u)^p]. Sampling only u on the sphere and inserting the exact radial moment is a valid Rao-Blackwellisation. However the radius is already highly concentrated at n=1024. The coefficient of variation of R^p is approximately 2.2%, 4.4%, 6.6%, and 8.8% for p=1,2,3,4 respectively. Removing only this source of noise cannot plausibly deliver a 100x reduction unless the angular observable is almost deterministic. It is still a free improvement for a sampling probe, but not a production strategy by itself.

2. The only sampling CV worth testing: an affine Gaussian surrogate

Use the same Gaussian input x to run the real network and a linearized surrogate whose ReLU is replaced by its first Hermite coefficient around the chain state. The surrogate remains affine in x, so its moments are exactly known. For each neuron and layer, use the centered surrogate powers as control variates for the actual centered powers.

> h_tilde_l = m_l + D_l W_l (h_tilde_{l-1} - m_{l-1}), D_l = diag(Phi_l).
>
> Controls: (z_tilde-mu)^2, (z_tilde-mu)^3, (z_tilde-mu)^4 with exact Gaussian expectations.

This control is mathematically clean; the problem is correlation. The previous public P2 experiments obtained only order-1.5x variance reduction from realizable MC controls for final outputs. Moment corrections could correlate better, so measure rather than assume.

Falsifier: run 8192 paired samples. Cross-fit the CV coefficient on half and evaluate on half. For the variance correction, kill unless the squared-observable control has R^2 >= 0.99 on the neurons that matter. For kappa_4, kill unless the fourth-moment/cumulant residual has R^2 >= 0.90-0.95. If those thresholds are not met, no 10^4-10^5 sample budget will be score-positive.

3. Antithetic, Sobol, and sphere QMC

Use them only on top of the paired-CV probe. Bias-free ReLU is homogeneous but not odd, so x and -x do not cancel the dominant nonlinear fluctuation. Low-discrepancy designs on a 1023-dimensional sphere also face high effective dimension after 16 random dense layers. Their job is to provide a modest extra factor, not the missing two orders of magnitude.

Q4 - Which experiment first?

The score arithmetic strongly favors cheap analytic corrections. Starting from raw MSE 2.290e-8 and cost 260 u, a full variance oracle would remain score-improving even if it cost roughly 228 extra units. A full K4 oracle breaks even at roughly 112 extra units. Even recovering only half of each oracle gain permits about 79 extra units for variance or 46 extra units for K4 before the score becomes worse than baseline.

Rank	Experiment	Expected extra cost	Why first / stop condition
1	Teacher-forced bivariate covariance super-closure, then port cost-4/5	Diagnostic: numpy n^2 only. Port: about 1-3 u/network	Highest oracle value; no new state; stop if projected dvar RMS improves <25%.
2	1111 directional K4 birth: star + path + 3+2(t_rep), last 6 layers	~7-10 u for m=300-400	Uses exact reference topology; no K4 memory; stop if K4 residual RMS improves <25%.
3	Add nine cheap 211 hypertree classes	~additional 5-15 u in a late-layer implementation, depending batching	Targets the measured missing pattern directly; stop if total K4 residual reduction <40%.
4	8192-sample paired affine-surrogate CV probe	~131 u for real samples, more if surrogate not fused	Run only to measure R^2. Production requires ~0.99 R^2 for variance.

Recommended one-hour experiments

11. Variance oracle ladder: on each stored layer evaluate V29 closure with true pair slices, then cost-4 and cost-5 pair super-closures. Plot fresh C residual RMS and projected dvar RMS. This is the single most informative experiment.
12. K4 1111 directional oracle: for 128 next-layer rows compute star and path exactly from C; add 3+2 first with t_rep, then with exact factorized-K3 contraction. Compare against the true missing K4-diagonal residual.
13. If the 1111 predictor works, implement the weighted 211 hypertree evaluator from the existing TERM_TABLE_211 graph list, but drop the three non-hypertree classes. Measure whether the cheap nine classes explain most of the residual.
14. Only after the analytic probes: run 8192 paired real/surrogate samples and measure CV R^2. Do not invest in a 10k-50k production sampler unless the required R^2 threshold is already met.

What I expect

My highest prior is that the variance channel can be improved cheaply because it is a strictly bivariate closure problem and the measured fresh error is at the size expected from one omitted diagram rung. For kappa_4, I expect the pure covariance star/path terms plus the repeated-slice approximation to the 3+2 term to explain a useful but not complete fraction of the missing per-neuron pattern. The weighted 211 hypertree terms are the next logical addition. I have a low prior that sampling can meet the required variance reduction at this width and depth.

I do not claim that these steps guarantee the 1.64e-8 frontier. They do, however, give three sharp falsifiers that can tell you within one network whether the remaining error is a cheap local-closure problem or whether genuinely new higher-order state is unavoidable.

Implementation sketches

A. Batched 1111 directional birth

```
# Q has shape (n,m): columns are selected rows of the next W, transposed
X = a1[:,None] * Q
R = C_off @ X                       # m/n u
STAR = 4 * colsum((a3[:,None]*Q) * R**3)
S = (a2[:,None]*Q) * R
PATH = 12 * colsum(S * (C_off @ S))       # m/n u
Trep = D21.T @ (X*X) + 2*X*(D21 @ X) - 2*D3[:,None]*(X*X)   # 2m/n u
K3C = 12 * colsum(S * Trep)
delta1111 = STAR + PATH + K3C.
```

The column sums produce one scalar per selected next-layer neuron. All temporary arrays are n by m. No fourth-order state is stored.

B. Weighted 211 hypertree evaluator

For each selected q: u=q*q, v=q.

For each leading TERM_TABLE_211 graph whose support hypergraph is a hypertree:

attach u * w(k,2) to the repeated axis a;

attach v * w(k,1) to axes b,c;

use C / C^2 / D21 / D21^T as edge messages;

eliminate leaves by matrix-vector products and multiply the resulting root messages;

subtract a=b and a=c coincidences exactly (same logic as eval_capped).

Multiply the final birth slice contraction by 6 and add to the next pre-activation K4 diagonal.

C. Variance super-closure generator

Reuse ARC's vector-partition combinatorics, but request only output partition (1,1), increase the diagram budget, and reject any input block whose total degree exceeds four. Compile the surviving records into the same fused elementwise program style used by V29. Because the output has only two axes, every factor is a vector or matrix and every term is O(n^2).

Evidence used and caveats

- The path/star/3+2 coefficients 12/4/12 are not guessed: they are explicitly present in ARC's public factor_k4.py implementation for the all-distinct fourth-cumulant birth.
- The V29 code already includes a memoryless K4 regeneration and an explicit (3,1) use-side term wk431, so the proposal above does not count that term as new. The target is the directional 211/1111 birth content that V29 does not explicitly carry.
- The public reference repository currently notes that factorized augmented algorithms omit non-hypertree top-slice diagrams of the same asymptotic squared order as some retained augment terms. That is why the 211 proposal separates the nine cheap hypertree classes from the three hard non-hypertree classes instead of assuming the cheap subset is exact.
- All score-gain calculations use the user-supplied Round-7 oracle MSEs and 260 u baseline cost. They are break-even calculations, not predictions of achieved leaderboard performance.

References

Wilson Wu, Victor Lecomte, Michael Winer, George Robinson, Jacob Hilton, Paul Christiano. "Estimating the expected output of wide random MLPs more efficiently than sampling." arXiv:2605.05179 (2026). https://arxiv.org/abs/2605.05179

Alignment Research Center public implementation: alignment-research-center/mlp_cumulant_propagation, especially src/mlp_kprop/factor_k3.py, factor_k4.py, and kprop_harmonic.py. Inspected 23 Sep 2026. https://github.com/alignment-research-center/mlp_cumulant_propagation

504aldo public Phase-2 implementation and research log: 504aldo/whest-p2-cumulant-k3, especially estimators/estimator_v29.py, lean/lean_k3_aug.py, docs/findings_log.md and docs/claims_evidence.md. Commit inspected: 1a4083f89184d6b1cc55d2e55a5a561d2ae50a8d. https://github.com/504aldo/whest-p2-cumulant-k3